

CS 409 - Final Project Proposal

Project Name: BookSwap Store

Team Name: Swapify 

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Motivation and Problem Statement

At the University of Illinois at Urbana–Champaign (UIUC), textbook costs have long been one of the most significant financial burdens for students. At the end of each semester, many books are left unused in dorm rooms or storage, while new students have to buy the exact same textbooks again. This cycle is inefficient, wasteful, and prevents valuable learning resources from being reused effectively.

For students in majors such as **accounting, economics, and the social sciences**, this issue is particularly serious. Required textbooks are often updated frequently and sold at very high prices. For example, *Essentials Of Statistics For Business And Economics* costs **\$149.99** at the Illini Union Bookstore (with the e-book still priced at **\$94.99**), and core accounting textbooks like *Intermediate Accounting (LI) W/Connect* often exceed **\$250** each. A student taking three or four major courses can easily spend close to a thousand dollars per semester on textbooks. For many who rely on scholarships, loans, or part-time jobs, this cost represents a heavy financial strain and reduces equal access to education.

Although some students turn to rentals or digital editions, these options often come with restrictions—limited access periods, lack of annotation features, or compatibility issues. As a result, many students prefer secondhand books or exchanges as a practical and sustainable alternative. However, **existing platforms do not effectively meet the needs of UIUC students:**

- **Poor search experience:** Platforms like Facebook Marketplace or Reddit r/UIUC lack course-based search or version filters, forcing students to scroll through unrelated listings.
- **Low trust and safety:** There is no campus-specific verification, and listings often mix in non-UIUC users, leading to scams or unreliable communication.
- **Fragmented process:** Students must switch between multiple platforms (Reddit, Messenger, Venmo) to complete a single transaction.
- **Resource waste:** Without a unified campus system, many reusable textbooks never reenter circulation.

To address these issues, we propose **BookSwap UI**, a web application designed specifically for UIUC students to exchange and trade textbooks safely and efficiently. The platform will integrate course

tagging, map-based location search, and UIUC email verification to build a secure and trustworthy environment. Students can find textbooks by course code (e.g., ECON 203, CS 409) and arrange local meetups to reduce shipping costs and scam risks.

Beyond a marketplace, **BookSwap UI** aims to foster a culture of sharing and sustainability within the UIUC community. By encouraging textbook reuse and peer-to-peer collaboration, the platform seeks to reduce waste, lower costs, and promote an equitable and efficient exchange of knowledge and resources across campus. To further enhance trust, the platform will also include a user rating system that ensures safe and reliable exchanges among verified students.

Features and Interactions

To reduce textbook costs, promote resource reuse, and strengthen trust within the campus community, **BookSwap UI** will provide a set of core features and user interactions that make textbook exchange safe, efficient, and easy to use.

The platform follows heuristic usability principles and uses a responsive interface design so that all interactions are simple, clear, and consistent across both desktop and mobile devices.

1. User Registration and Campus Verification

- Students first create an account using their UIUC email address.
 - A verification code is sent to the email, and users must complete this verification before they can post textbooks or place orders.
 - Unverified users can still browse listings but cannot contact sellers or make purchase requests.
- The registered email is used **only for identity verification and is not displayed to other users**.
 - This ensures that only verified UIUC students can access full platform features while protecting user privacy and preventing fake accounts or scams.

2. Textbook Posting and Management

- Students can post textbooks through a simple form that includes the title, course code, author, edition, price, and meeting location.
 - Users can upload textbook covers and notes to help buyers confirm the correct version and condition.
- Listings can be edited, removed, or updated before receiving an offer, allowing users to manage their posts more effectively.

3. Search by Course Code and Keywords

- The search function allows users to find textbooks by course code (for example, ECON 203 or CS 409), title, author, or keywords.
 - Results can be sorted by price, distance, or posting time to help students quickly locate what they need.

4. Textbook Details and Purchase Requests

- Each listing includes detailed information such as course tags, edition, ISBN, seller information, and book status.
- Buyers can click a “Send Purchase Request” button to contact the seller.

- The seller will receive a notification and can view, accept, or decline the request in the message center.
- 5. **User rating**
 - After each completed transaction, users can rate each other on a 1–5 star scale and optionally leave brief comments about the interaction.
 - Average ratings will appear on each user’s profile and listing pages to help others assess reliability.
 - New users will start with a neutral rating, and suspicious or consistently low-rated accounts may be flagged for review by administrators.
 - To prevent abuse:
 - Only verified users who have completed a transaction can leave a rating.
 - Ratings are anonymous and non-editable once submitted.
 - Admins can remove false or inappropriate reviews.
- 6. **Map View and Location Filter (Optional)**
 - Depending on project progress, a map view may be added later.
 - It will allow students to see nearby listings and choose in-person exchanges within the UIUC campus.
- 7. **Admin Management**
 - Administrators have access to simple management tools that allow them to edit the list of available courses, update course tags, and maintain platform data consistency.

Competitive Analysis

Our main competitors include the **Illini Union Bookstore (IUB)**, **Facebook Marketplace**, **Reddit r/UIUC**, **Chegg / BookScouter**, and the **Textbook Exchange Network (TEN)**.

- **Illini Union Bookstore (IUB)**
The official UIUC bookstore offers in-person buyback and rental services at the end of each semester.
 - **Strengths:** trusted institution, direct on-campus service.
 - **Limitations:** fixed pricing, no student-to-student exchange, no online peer listings.
- **Facebook Marketplace**
A general-purpose secondhand marketplace where many students buy and sell textbooks informally.
 - **Strengths:** wide reach, flexible negotiation, fast posting.
 - **Limitations:** no campus filtering, public user profiles, high scam potential.
- **Reddit r/UIUC**
A local online community where students can post textbook requests or offers in forum-style threads.
 - **Strengths:** familiar to students, socially driven, fully free.
 - **Limitations:** no structured listings, poor searchability, no safety mechanisms.

- **Chegg / BookScouter**

Commercial textbook platforms that offer rental, buyback, and price comparisons for textbooks across vendors.

- **Strengths:** professional inventory, national coverage, price transparency.
- **Limitations:** lacks local exchange, no UIUC course integration, publisher-controlled pricing.

- **Textbook Exchange Network (TEN)**

A non-profit peer-to-peer textbook exchange system available at some U.S. colleges.

- **Strengths:** student-focused, supports direct exchange.
- **Limitations:** not available at UIUC, lacks integration with course data, minimal customization.

The table below summarizes the key differences in feature support across these platforms compared to the BookSwap UI's proposed features:

Feature	BookSwap UI	Illini Union Bookstore (IUB)	Facebook Marketplace	Reddit r/UIUC	Chegg / BookScouter	Textbook Exchange Network (TEN)
UIUC Student Verification	Yes	Optional	No	No	No	Optional
Course-Based Search / Tags	Yes	No	No	No	No	No
Peer-to-Peer Exchange	Yes	No	Yes	Yes	No	Yes
On-Campus Transactions	Yes	In-person only	Yes	Yes	Shipping only	Partial
Buyback or Direct Selling Price Control	Negotiable between users	Fixed by store	Negotiable	Negotiable	Publisher controlled	Negotiable
Search and Filtering Tools	Advanced	Manually by staff	Keyword only	Manual browsing	ISBN/keyword search only	Keyword only
Map View Support	Yes (planned)	No	Yes	No	No	No
Safety / Scam Prevention	Verified users	Verified by staff	No	No	Account-based reputation	Account-based reputation
Available On	Responsive Web App	Physical	Website + app	Website + app	Website + app	Website